Building Our Digital World: Computer Systems and Architecture COMP1860

School of Computer Science, University of Leeds

Activity Sheet 1.4

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio. Portfolio activities are detailed in the section "Portfolio Task". Activities marked by (*) are advanced activities and may take some time to complete.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided. Activities marked as part of the portfolio should be submitted to Gradescope before the completion date (Friday, 30/10/2026, 14:00).
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
3. **Integrity:** You are responsible that the evidence you submit as part of your portfolio evidence is entirely your own work. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on activities submitted as part of the portfolio will be available on Gradescope..

Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via the online booking system.

Expected time for completion: 2-3 hours.

Expected complete date: Friday, 30/10/2026, 14:00

Coursework summary

This worksheet complements week 4's material. It covers minimisation of Boolean expressions using the two methods covered in the lectures: Karnaugh maps and Boolean algebraic laws. You should refer to lectures 1.4.1 and 1.4.2 when attempting these questions.

Learning outcomes

On completion of this activity sheet you will:

1. Apply minimisation of Boolean expressions to simplify logic circuits.

Portfolio Task

The portfolio task is a Gradescope online quiz. You will find the link to it in the COMP1860 Minerva module.

Formative Questions

1. Construct a truth table for each of the formulae below:
   (a) $(a \vee b) \wedge (a \vee c)$
   (b) $\neg(\neg a \vee \neg b \vee \neg c)$
   (c) $(a \vee b) \wedge (a \vee \neg c)$
   (d) $(a \wedge \neg c) \vee (b \wedge c)$
   (e) $a \vee (\neg b \wedge \neg c) \vee (a \wedge b)$
   (f) $(a \wedge b \wedge c) \vee (a \wedge b \wedge \neg c)$
2. Apply Boolean algebraic laws to simplify each of the formulae below:
   (a) $(a \vee b) \wedge (a \vee c)$
   (b) $\neg(\neg a \vee \neg b \vee \neg c)$
   (c) $(a \vee b) \wedge (a \vee \neg c)$
   (d) $(a \wedge \neg c) \vee (b \wedge c)$
   (e) $a \vee (\neg b \wedge \neg c) \vee (a \wedge b)$
   (f) $(a \wedge b \wedge c) \vee (a \wedge b \wedge \neg c)$
3. Use a Karnaugh map to construct a minimised DNF formula for each of the expressions below:
   (a) $(a \vee b) \wedge (a \vee c)$
   (b) $\neg(\neg a \vee \neg b \vee \neg c)$
   (c) $(a \vee b) \wedge (a \vee \neg c)$
   (d) $(a \wedge \neg c) \vee (b \wedge c)$
   (e) $a \vee (\neg b \wedge \neg c) \vee (a \wedge b)$
   (f) $(a \wedge b \wedge c) \vee (a \wedge b \wedge \neg c)$
4. (*) Use a Karnaugh map to construct a minimised CNF formula for each of the expressions below:
   (a) $(a \vee b) \wedge (a \vee c)$
   (b) $\neg(\neg a \vee \neg b \vee \neg c)$
   (c) $(a \vee b) \wedge (a \vee \neg c)$
   (d) $(a \wedge \neg c) \vee (b \wedge c)$
   (e) $a \vee (\neg b \wedge \neg c) \vee (a \wedge b)$
   (f) $(a \wedge b \wedge c) \vee (a \wedge b \wedge \neg c)$
5. (*) For each of the formulae below:
   - apply Boolean algebraic laws to simplify the Boolean expression;
   - construct the truth table;
   - use a Karnaugh map to construct a minimised DNF formula;
   - use a Karnaugh map to construct a minimised CNF formula.

   (a) $(a \wedge b \wedge c) \vee (c \wedge d) \vee (\neg a \wedge (b \vee d)) \vee (d \wedge (\neg a \vee c))$
   (b) $(d \wedge (a \vee b)) \vee (\neg b \wedge ((c \wedge \neg d) \vee (\neg c \wedge \neg d)))$